MILITARY SPECIFICATION

STOPCOCK, SURGICAL SYRINGE, ONE WAY, TUBING ATTACHMENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a metallic one-way stopcock suitable for use in intravenous therapy.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical; General Specifications.
PPP-B-566	Box, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Boxes, Shipping, Fiberboard.
PPP-B-676	Boxes, Setup.

Military

MIL-P-116	Preservation - Packaging, Methods of.
MIL-L-10547	Liner, Case and Sheet, Overwrap; Water Vaporproof or Waterproof, Flexible.

FSC 6515

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications, standards, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z70.1	American Standard Dimensions of Glass and Metal Luer Tapers for Medical Applications.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, NY 10018.)

3. REQUIREMENTS

3.1 Material. The entire stopcock shall be fabricated from Class 8, Type 260, 270 or 360 material in accordance with GG-I-526. All surfaces shall be chromium plated over nickel plated, except the bearing surfaces of the stopcock body, rotating key and washer, and the bore through the rotating key. As an alternate, the surfaces specified to be plated shall be nickel plated only. The boil test and plating requirements specified in GG-I-526 shall apply.

3.2 Design and construction. The item shall be a one-way stopcock with a female Luer lock connector in accordance with Figure 3 of ANSI Standard Z70.1 at one end and a connector for 1/16 inch inside diameter rubber tubing at the other end. The connections of the stopcock shall form secure and leak-proof junctions when a Luer slip or Luer lock syringe and 1/16 inch inside diameter rubber tubing are assembled to the stopcock. The key shall be easily rotatable in both directions.

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3.3 Diameter and depth of female Luer lock cavity. The cavity of the female Luer lock connector shall be tested for diameter and depth of Luer cavity as specified in 4.4.1. Cavities which do not fit the plug gage within the "GO"-"NO GO" limits specified in 4.4.1 shall be rejected. In addition, when the cavity is fitted onto the plug gage, there shall be no movement of the cavity on the gage when connector is pressed lightly from side to side in all directions.

3.4 Leakage of stopcock. When tested as specified in 4.4.2 the female Luer lock connector, the tubing junction, and the junction of the rotating key shall not show leakage exceeding the formation of one clinging drop. The rotating key shall not separate or loosen from the stopcock body during the leakage test.

3.5 Clear passage of fluid path. The entire fluid path of the stopcock shall be clear and free from foreign particles to allow free passage of fluid as specified in 4.4.3.

3.6 Security of assembly. When tested as specified in 4.4.4, the rotating key shall not pull out of, or separate from, the stopcock body.

3.7 Identification marking. The stopcock shall be marked with the name or registered trademark of the contractor. Stopcocks of other than U.S. manufacture shall also be marked with the country of origin.

3.8 Workmanship. Workmanship shall be first class throughout. Stopcocks shall be free from burrs, sharp edges, cracks, holes, pits, scale, corrosion, flaking or incomplete plating, foreign particles, and any other defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality supplied by the manufacturer of the copper alloys may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number (NSN), the item identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, the individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and as indicated in Table I. The unit of product for examination purposes shall be one stopcock.

Table I. Sampling for examination

Type defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in Table II. The unit of product for test purposes shall be one stopcock. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

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Table II. Sampling for tests

Component	Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Stopcock	Boil test	3.1	GG-I-526	S-2	1.5
	Plating test	3.1	GG-I-526	S-2	1.5
	Diameter and depth of female Luer cavity	3.3	4.4.1	S-4	1.0
	Leakage test	3.4	4.4.2	S-4	1.0
	Clear passage of fluid test	3.5	4.4.3	S-2	1.5
	Security of assembly	3.6	4.4.4	S-2	1.5

4.2.3 Raw materials. A sufficient quantity of each lot of each metal used in the contruction of the stopcock shall be selected for chemical analysis.

4.3 Examination.

4.3.1 Examination of stopcock. The stopcock shall be examined to determine compliance with the requirements contained in this specification. Examination shall be conducted as specified in Tables I and III. Examination, however, shall not be restricted to defects listed in Table III. Any stopcock in the sample containing one or more defects shall be rejected and if the number of defective stopcocks in any sample exceeds the acceptance number for that sample, the lot represented by that sample shall be rejected.

Table III. Classification of defects

Categories	Defects
Critical	None defined.
Major	
101	Stopcock not of design specified.
102	Rotating spigot (key) does not exhibit smooth operation, from the open to close position, without sticking or binding in any position.
103	Stopcock not plated as specified.
104	Plating not uniform.
105	Plating not free from flaking.
106	Fluid paths not free from foreign matter and obstructions.

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Table III. Classification of defects (cont'd)

Categories	Defects
107	Stopcock not free from burrs, sharp edges, cracks, holes, pits, scale or corrosion.
108	Rotating spigot not secured to stopcock body (i.e. separates from body when rotated).
Minor	
201	Stopcock not identified as specified in 3.7.

4.3.2 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

Table IV. Examination of preparation for delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

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4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Diameter and depth of female Luer lock cavity. The cavity of the female Luer lock connector shall be tested with the male Luer taper plug gage in accordance with Figure 5 of ANSI Standard Z70.1 (this gage shall be as shown in the ANSI Standard, except that it shall be constructed with a suitable fluid path to permit performance of the leakage test specified in 4.4.2). The female Luer cavity shall be placed onto this gage with a straight motion, without turning or twisting the plug gage into the cavity. Once the female Luer cavity is placed onto the gage, no additional force shall be exerted to force the plug gage into the Luer cavity. The fit of the Luer cavity onto this gage shall be tested by means of the feeler gage in accordance with Figures 5 and 6 of ANSI Standard Z70.1.

4.4.2 Leakage test. The stopcock shall be subjected to an internal water pressure of 45 pounds per square inch gage (psig). The plug gage specified in 4.4.1 shall be inserted into the female Luer cavity with a straight motion, without turning or twisting the plug gage into the cavity with a force not to exceed 5 pounds. The feeler gage shall be placed onto the plug gage to assure that the "GO" position is not violated. Tubing with a 1/16 inch nominal inside diameter shall be placed onto the tubing connection of the stopcock and sealed in a leakproof manner. Water shall then be internally admitted to the testing arrangement in a leakproof manner until the entire fluid path of the stopcock is filled (air bubbles eliminated) with internal pressure of 45 psig for 30 seconds. The entire assembly shall be noted for leakage and spigot performance as specified in 3.4.

4.4.3 Clear passage of fluid path. The stopcock shall be tested with an appropriate testing fluid applied internally to the stopcock with the stopcock placed in a horizontal position. The entire fluid path shall be tested using the testing assembly fixtures specified in 4.4.2 except that the end of the stopcock shall be opened to the atmosphere for fluid exit. The stopcock shall allow free passage of the testing fluid within 5 seconds after an internal pressure of 3 $\pm$ 1/2 psig has been applied to the stopcock. One liter of testing fluid shall be passed through the stopcock and then through filter paper. The filter paper shall be examined for presence of foreign matter.

4.4.4 Security of assembly. The stopcock shall be tested for security of assembly by applying a straight force or pull of 10 pounds to the rotating spigot member for a period of at least 1 minute with the stopcock body appropriately secured.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit package. Each stopcock shall be packaged in accordance with MIL-P-116, method IC-1.

5.1.1.2 Intermediate package. Twelve (12) stopcocks shall be packaged in a box of appropriate size and design constructed in accordance with PPP-B-566 or PPP-B-676. Closure shall be adequate to prevent accidental opening under normal handling.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).

5.2.1 Level A. Seven hundred and twenty (720) stopcocks shall be packed in a box designed for a Type 2 load and constructed in accordance with PPP-B-585, Class 3, Style 3, PPP-B-601, overseas type; PPP-B-621, Class 2; or PPP-B-636, Class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof in accordance with PPP-B-636.

5.2.2 Level B. Seven hundred and twenty (720) stopcocks shall be packed in a box designed for a Type 2 load and constructed in accordance with PPP-B-585, Class 1, Style 3; PPP-B-601, domestic type; PPP-B-621, Class 1; or PPP-B-636, Class domestic. Closure shall be as specified in the appendix of the applicable box specification.

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5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contract shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The stopcock covered by this specification is intended to be used in intravenous therapy. One end should be capable of being attached to a 1/16 inch inside diameter rubber tubing and the other end to a Luer slip or a Luer lock syringe.

6.2 Ordering data. Procurement documents should specify:

(a) Title, number, and date of this specification.
(b) National Stock Number (NSN)
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification is intended to cover the following item appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6515-00-374-4700	STOPCOCK, SURGICAL SYRINGE, ONE WAY, TUBING ATTACHMENT

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1899
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-S-37121 STOPCOCK, SURGICAL SYRINGE, ONE WAY, TUBING ATTACH

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

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